BNCHMRK FRTRN90 FILE NAME/TYPE= TESTIN/TEST4.F90
BNCHMRK FRTRN90
BNCHMRK FRTRN90 CREATION DATE/TIME= 09-11-2026 16:05:46
BNCHMRK FRTRN90
BNCHMRK FRTRN90 FILE= 001 PAGES= 0002 LINES= 000082
BNCHMRK FRTRN90
BNCHMRK FRTRN90 SYSTEM= LINUX(7.1.5+KALI-AMD64)
BNCHMRK FRTRN90
BNCHMRK FRTRN90 SYSID= ACID SYSUSER= ACID
BNCHMRK FRTRN90
BNCHMRK FRTRN90 FORM= MEDIUM
BNCHMRK FRTRN90
BNCHMRK FRTRN90 CHAR= FONTMONO
BNCHMRK FRTRN90
BNCHMRK FRTRN90 PRT1403 VERSION= 1.6_PRE-RELEASE

```
                FFFFFFFFF RRRRRRRR  TTTTTTTT  RRRRRRRR  N       NN  9999999    00000
                FFFFFFFFF RRRRRRRRR TTTTTTTT  RRRRRRRRR NN      NN 999999999  0000000
                FF        RR     RR    TT     RR     RR NNN     NN 99     99 00     00
                FF        RR     RR    TT     RR     RR NNNN    NN 99     99 00     00
                FFFFFFFFF RRRRRRRRR    TT     RRRRRRRRR NN NN   NN 999999999 00     00
                FFFFFFFFF RRRRRRRR     TT     RRRRRRRR  NN  NN NN  99999999 00     00
                FF        RR  RR       TT     RR  RR    NN   NNNN        99 00     00
                FF        RR   RR      TT     RR   RR   NN    NNN        99 00     00
                FF        RR    RR     TT     RR    RR  NN     NN  99999999  0000000
                FF        RR     RR    TT     RR     RR NN      N  9999999    00000



                BBBBBBBB  N       NN  CCCCCCC  HH     HH M       M RRRRRRRR  KK     KK
                BBBBBBBBB NN      NN CCCCCCCCC HH     HH MM     MM RRRRRRRRR KK    KK
                BB     BB NNN     NN CC     CC HH     HH MMM   MMM RR     RR KK   KK
                BB     BB NNNN    NN CC        HH     HH MMMM MMMM RR     RR KK  KK
                BBBBBBBB  NN NN   NN CC        HHHHHHHHH MM MMM MM RRRRRRRRR KKKKK
                BBBBBBBB  NN  NN  NN CC        HHHHHHHHH MM  M  MM RRRRRRRR  KK  KK
                BB     BB NN   NNNN CC        HH     HH MM     MM RR  RR    KK   KK
                BB     BB NN    NNN CC     CC HH     HH MM     MM RR   RR   KK    KK
                BBBBBBBBB NN      NN CCCCCCCCC HH     HH MM     MM RR    RR  KK     KK
                BBBBBBBB  NN       N  CCCCCCC  HH     HH MM     MM RR     RR KK     KK



                                    00000     00000       1
                                   0000000   0000000     11
                                  00     00 00     00   111
                                  00     00 00     00    11
                                  00     00 00     00    11
                                  00     00 00     00    11
                                  00     00 00     00    11
                                  00     00 00     00    11
                                   0000000   0000000   111111
                                    00000     00000    111111
```

```
program lid_driven_cavity
  implicit none
  integer, parameter :: N = 50   ! grid size (NxN grid)
  real :: dx, dy, dt, Re         ! grid spacing, time step, Reynolds number
  real :: u(N, N), v(N, N), p(N, N)  ! velocity and pressure fields
  integer :: i, j, step
  real :: start_time, end_time, elapsed_time

  ! Parameters
  dx = 1.0 / (N-1)   ! Grid spacing in x direction
  dy = 1.0 / (N-1)   ! Grid spacing in y direction
  dt = 0.001         ! Time step size
  Re = 100           ! Reynolds number

  ! Initialize arrays
  u = 0.0
  v = 0.0
  p = 0.0

  ! Initialize the top boundary (lid) velocity
  u(N, :) = 1.0

  ! Start timing
  call cpu_time(start_time)

  ! Main loop for time stepping
  do step = 1, 1000
     call compute_velocity(u, v, p, dx, dy, dt, Re)
     call update_pressure(p, dx, dy)

     ! Output or check convergence
     if (mod(step, 100) == 0) then
        print *, 'Step: ', step
     end if
  end do

  ! Stop timing
  call cpu_time(end_time)
  elapsed_time = end_time - start_time
  print *, 'Elapsed time for CFD simulation: ', elapsed_time, ' seconds'

contains

  ! Function to update the velocity and pressure fields (simplified)
  subroutine compute_velocity(u, v, p, dx, dy, dt, Re)
    real, dimension(:,:), intent(inout) :: u, v, p
    real, intent(in) :: dx, dy, dt, Re
    integer :: i, j

    ! Simple explicit method for velocity (simplified)
    do i = 2, N-1
        do j = 2, N-1
            u(i, j) = u(i, j) - dt * ( (u(i, j) * (u(i+1, j) - u(i-1, j))) / (2*dx) + &
                                       (v(i, j) * (u(i, j+1) - u(i, j-1))) / (2*dy) )
        end do
    end do

    ! Simple velocity update for v (similar)
    do i = 2, N-1
        do j = 2, N-1
```

```
            v(i, j) = v(i, j) - dt * ( (u(i, j) * (v(i+1, j) - v(i-1, j))) / (2*dx) + &
                                       (v(i, j) * (v(i, j+1) - v(i, j-1))) / (2*dy) )
        end do
    end do
  end subroutine compute_velocity

  ! Function to solve for pressure (simplified Poisson equation solver)
  subroutine update_pressure(p, dx, dy)
    real, dimension(:,:), intent(inout) :: p
    real, intent(in) :: dx, dy
    integer :: i, j

    ! Simple pressure Poisson equation (Jacobi iteration)
    do i = 2, N-1
        do j = 2, N-1
            p(i, j) = 0.25 * ( p(i+1, j) + p(i-1, j) + p(i, j+1) + p(i, j-1) )
        end do
    end do
  end subroutine update_pressure

end program lid_driven_cavity
```